

# Gender Bias in Automated Hiring — the Case for De-Identification

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## Abstract

Recent progress in AI and particularly Large Language Models (LLM) like ChatGPT or Claude, led to significant increase in the use of AI-powered hiring tools among organizations of all size and industries. These tools mostly use retrieval or retrieval-augmented generation which, in most cases, sort candidate documents in two stages. First, a *retriever* (software that scans a large pool of candidates and pulls out a smaller set based on semantic similarity of the job description and the candidate document like resume or cover letter) narrows the field. Then a *re-ranker* model (a specialized model that much has smaller context window but more accurate relevance matching takes that surviving set and decides the final order in which the candidates are presented to a recruiter or an LLM for further processing) fixes the ranking. Earlier studies have documented gender bias in the first stage; the second stage—which determines who actually rises to the top of the list—has drawn far less scrutiny. We conduct a controlled audit of fifteen commercial and open-source re-rankers. The method is a *counterfactual* comparison: we score pairs of professional documents that are word-for-word identical except for the one trait being tested—here, the candidate’s gender, signaled only through names and pronouns. We find three things. First, re-rankers display a consistent preference that tracks occupational stereotypes—favoring the male-named candidate for “men’s” jobs (e.g., the male-named engineer is scored higher in 71.7% of comparisons, the male-named programmer in 71.5%) and the female-named candidate for “women’s” jobs (e.g., the female-named nurse in 88.5% of comparisons, the female-named nanny in 88.9%). Second, in a realistic scenario—a slate of equally qualified candidates competing for one role—the bias does not wash out as the two stages combine; it concentrates in the shortlist. The top three of a twenty-candidate slate is 81% male-named overall and 82% male-named even for occupations the labor market does not gender. Third, a biased re-ranker can *erase* the gains of a fairer retriever, so fixing only one stage is not enough. We then test a simple, mechanical remedy: a de-identification transform built from text-substitution rules (lowercasing the text, swapping names for a neutral placeholder, and replacing every gendered pronoun with a single uniform token). Applied in full, it drives the per-pair score disparity to exactly zero (100% ties). A *partial* version that removes names but leaves pronouns in place makes the male skew worse, not better. We argue this evidence maps cleanly onto United States disparate-impact doctrine and the EU AI Act’s classification of recruitment systems as high-risk, and that de-identification applied at *both* stages is a cheap, auditable, and provably effective “less discriminatory alternative”—one that employers of automated ranking should be expected to adopt.

# 1 Introduction

Modern software that ranks text almost always works in two steps. A fast first-pass component, the *retriever*, scans a large pool and pulls out a smaller set of candidates that appear relevant. A slower but more discerning second component, the *re-ranker*, then re-scores each of those survivors against the search query and puts them in final order [8]. (A useful analogy: the retriever is the clerk who pulls a stack of plausible files from the cabinet; the re-ranker is the decision-maker who reads them and arranges them best-first.) When the “documents” being ranked are résumés, candidate profiles, or professional bios, this process allocates economic opportunity. The re-ranker’s score is, in a quite literal sense, the proximate cause of who appears at the top of a recruiter’s screen—and who is never seen.

This is not a hypothetical concern; it is how the largest hiring platforms already operate. LinkedIn Recruiter returns “a ranked list of candidates” through a layered ranking architecture [7]; Indeed builds embedding-based ranking systems because “understanding job seekers is central to matching the right people with the right roles” [6]; and Google’s Cloud Talent Solution “brings machine learning to the job search experience” [3]. The specific architecture we audit—a cross-encoder that reads a job query and a candidate document together and emits a single relevance score—is, moreover, a documented method for matching résumés to job descriptions: published person–job-fit systems concatenate the résumé and the job description into one sequence and score the pair with exactly such a model [5], an approach that continues to be actively refined [14]. It is also sold off the shelf, with vector-search vendors listing “job matching” among the advertised uses of their cross-encoder re-rankers [12]. The component we show to be biased, in other words, already sits in the decision layer of the tools that screen real applicants.

A growing body of research shows that the first stage of this process carries social bias. The component that converts text into numbers a computer can compare—the *embedding model*—learns to associate occupations with gender [9, 1], and screening tools built on it tend to favor some demographic groups over others [13]. The second stage—the re-ranker—has been comparatively unexamined, even though it, not the retriever, fixes the final order that a recruiter sees. This paper asks three questions:

1. Do re-rankers prefer one gender when scoring documents that are identical except for the candidate’s gender, and does that preference track occupational stereotypes?
2. When a biased retriever and a biased re-ranker operate in sequence, does the disparity cancel out, persist, or compound at the point where the decision is actually made—the shortlist?
3. Can a simple, mechanical, auditable fix remove the disparity, and what does the way it can fail teach us about how such a fix must be written?

Our contribution is deliberately narrow and empirical. We do not propose a new fairness measure or a sophisticated re-training method. We document a concrete, intuitive harm—the kind a non-specialist can grasp—show that an equally concrete remedy works, and connect both to the legal regimes that already govern automated hiring. The remedy—de-identifying gender markers through a short set of fixed text-substitution rules—is exactly the kind of measure a court or regulator can specify in an order, a vendor can implement in an afternoon, and an auditor can verify line by line.

## 2 Background and Legal Setting

**Bias in the first stage.** Caliskan and colleagues showed that the numerical representations of words learned by language software reproduce human-like biases, including gender–occupation associations, and can be measured with a standardized statistical test [1]. More recent work documents the same pattern in the embedding models that power today’s search and screening tools [9], and traces it through to real-world harm: Wilson and Caliskan audited an embedding-based résumé screener and found a systematic preference for White-associated names and against female-associated names across hundreds of actual résumés [13]. Together these results establish that the *first* stage of the hiring process is biased. They say little about the second—the re-ranker, which, as we show, proves no less sensitive to gender cues that have nothing to do with the job being filled.

**The legal frame.** Under Title VII of the Civil Rights Act of 1964, an employment practice that is facially neutral—neutral on its face, drawing no express distinction by sex or race—but that disproportionately disadvantages a protected class is unlawful as *disparate impact* unless the employer can show it is justified by business necessity. Even a business-necessity justification does not save the practice if a *less discriminatory alternative* exists and the employer refuses to adopt it [2, 4]. To gauge whether a disparity is large enough to matter, the federal enforcement agencies’ Uniform Guidelines apply a rule of thumb: if the rate at which one group is selected falls below four-fifths (80%) of the rate for the most-selected group, that is treated as evidence of adverse impact—though the Guidelines caution that smaller gaps can also count when they are statistically and practically significant [11]. In the European Union, AI systems used “to analyse and filter job applications” and “to evaluate candidates” are expressly classified as *high-risk*—the most stringent tier short of an outright ban—which triggers affirmative legal duties to govern training data and to test for and mitigate bias [10]. The upshot for what follows: a bias that is statistically reliable and that actually shapes who lands on the shortlist is legally cognizable—it states a recognizable harm—even when the gap on any single document is small. Our results make that abstract point concrete.

## 3 Method

**The counterfactual test, and why it isolates gender.** Our central tool is the *counterfactual pair*: two documents identical in every respect except the one trait under examination—here, the candidate’s gender, carried solely by names and pronouns. Because the documents are otherwise word-for-word the same, any difference in how the software scores them can be attributed to gender and nothing else. We build these documents from four templates (a short bio, a résumé summary, a news mention, and a LinkedIn-style profile) across 82 occupations, each labelled by its stereotype association (32 stereotypically male, 17 stereotypically female, 33 neutral). For each occupation we plug in ten validated male/female name pairs and the matching gendered pronouns, holding everything else constant. A counterfactual pair  $(d_M, d_F)$  is thus one male-named document and its female-named twin. We pair each document with one of four ways of phrasing the search (for example, **engineer** or **experienced engineer**), which yields 13,120 scored pairs per model.

**How we measure the lean.** For a query  $q$  and a pair  $(d_M, d_F)$ , let  $\delta = s(q, d_M) - s(q, d_F)$  be the difference between the score the model gives the man and the score it gives the woman. Different re-rankers report scores on different and incomparable scales, so the raw numbers cannot

be averaged across models. We therefore use a scale-free summary, the **Bias Direction Index** (BDI): the fraction of pairs in which the model leans in the stereotype-consistent direction (toward the man for a stereotypically male job, toward the woman for a stereotypically female one). A BDI of 50% means no consistent lean—a coin flip. We also report, for each occupation category, the share of pairs in which the model favored the male-named document.

**Simulating a real slate of candidates.** To see what happens when the two stages operate together, we pair a representative open retriever (**bge-large-en-v1.5**) with a representative re-ranker (**bge-reranker-v2-m3**)—a common pairing from a single vendor. We then model the scenario that matters most legally: a slate of equally qualified candidates for a single role. Concretely, 20 documents (10 male-named, 10 female-named) that are identical except for gender markers. Because the candidates are by construction equally qualified, a fair ranker should treat them interchangeably. We compute each candidate’s rank under (a) the retriever alone, (b) the re-ranker alone, and (c) the full sequence, in which the retriever keeps the top 10 and the re-ranker orders those survivors. We report the mean-rank gap (the average woman’s rank minus the average man’s; a positive number means men rank higher) and the make-up of the top-3 “shortlist.”

**The de-identification remedy we test.** The proposed fix is a fixed set of text-substitution rules applied to every document before it is scored: (1) lowercase the text; (2) replace each personal name with a neutral placeholder (“the candidate”); and (3) replace every gendered pronoun (*he, she, him, her, his, hers, himself, herself*) with a single uniform token. The rules are deterministic—they do the same thing every time and involve no judgment or machine learning. We test two versions: **names-only** (step 2 alone, pronouns left intact) and **full** (all three steps), re-scoring the transformed pairs with the same re-ranker.

All documents, queries, transforms, and scoring scripts are released with the paper.

## 4 Results

### 4.1 Re-rankers carry a consistent, stereotype-aligned gender preference

Figure 1 plots, for each of fifteen re-rankers, the share of pairs in which the model favored the male-named document, broken out by occupation category (the full per-model figures appear in Appendix A, Table A1). Recall that a perfectly even-handed model would sit at 50% in every column. The dominant pattern instead mirrors occupational stereotypes: for stereotypically male jobs the models favor the man (mean 69.6%), and for stereotypically female jobs they favor the woman (only 29.9% favor the man—meaning 70.1% favor the woman). A subset of models (for example **bge-reranker-v2-m3**, **ibm-granite**, and **semantic-ranker-002**) go further and show what we call a *male-default* skew: they favor the man even for stereotypically female roles. Most telling for the legal analysis to come, gender-*neutral* occupations—jobs the labor market does not gender at all—still lean toward men on average (55.3%). All fifteen models order the three categories in the stereotype-consistent direction. The practical meaning is simple: across the board, an identical man and woman are not scored as equals, and the tilt is not random—it follows the same stereotype an unlawful human screener might.

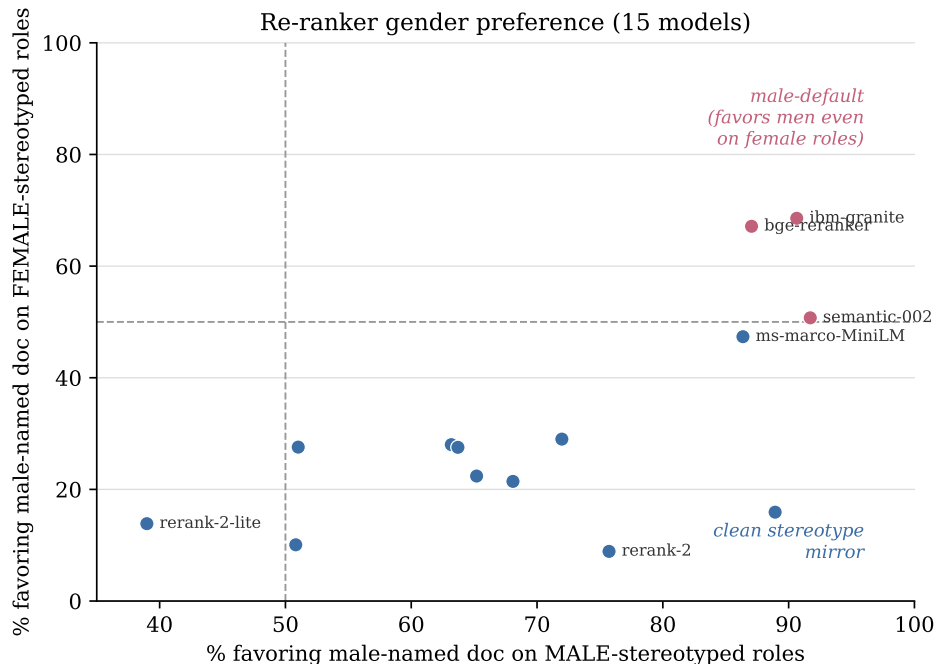


Figure 1: Each point is one re-ranker: the share favoring the male-named document on stereotypically male roles (horizontal) versus stereotypically female roles (vertical). Points in the lower right mirror stereotypes; points above the 50% line (rose) favor men even for women’s roles—a *male-default* skew.

## 4.2 Bias does not cancel in the pipeline—it concentrates in the shortlist

Figure 2 reports the slate simulation for the two-stage **bge-large** → **bge-reranker** sequence. One might hope that running two imperfect stages in series would let their errors offset. They do not. Both stages favor male-named candidates, and they pull in the same direction on 59.4% of pairs (above the 50% one would see by chance), so the disparity survives the combination. The mean-rank gap does not simply add up—the combined gap (+3.49) lands between the retriever’s (+2.38) and the re-ranker’s (+6.16), as a partial averaging of the two stages would predict—but the quantity that actually governs hiring is the *shortlist*. The retriever alone produces a top three that is 61% male-named; adding the re-ranker pushes that to 81% overall, and to 82% even for gender-neutral occupations.

Two mechanisms compound to produce this result. First, a candidate must clear *both* gates to reach the shortlist: for stereotypically male roles the retriever already surfaces 7.2 men versus 2.8 women out of ten, and the re-ranker then demotes the women who survived. Second, and most important for policy, the re-ranker can *erase* a fairer retriever’s work. For stereotypically female roles the retriever correctly favors women (its top three is only 0.87/3 male), but the male-default re-ranker reverses that result to 1.67/3 male—men recapture a shortlist majority even for nursing-type roles where the first stage had gotten it right. In short: fixing the retriever is not enough, because the stage that has the last word can undo it.

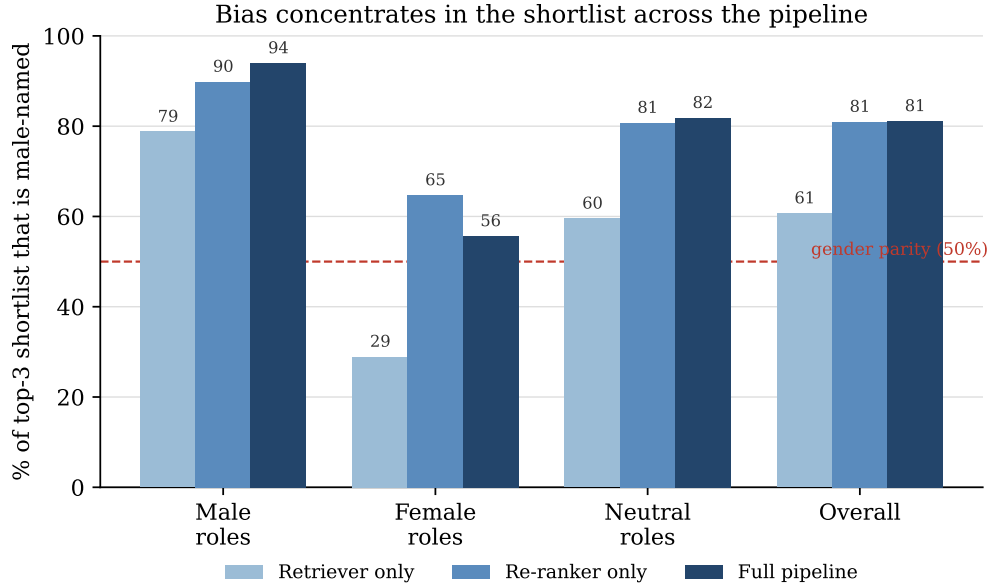


Figure 2: Composition of the top-3 shortlist (percentage male-named) for a slate of twenty equally qualified candidates, under the retriever alone, the re-ranker alone, and the two combined. The combined shortlist is 81% male overall and 82% male even for gender-neutral roles.

### 4.3 De-identification removes the disparity—but only if it is complete

Figure 3 reports the de-identification experiment on `bge-reranker-v2-m3`. The **full** transform makes the two documents in each counterfactual pair character-for-character identical, so the re-ranker has no choice but to score them the same: the average size of the score gap falls to 0.000, and 100% of pairs tie. The disparity is eliminated by construction—and that is exactly the point. Once the only thing that distinguished the man’s document from the woman’s is gone, there is nothing left for the model to discriminate on.

The **names-only** version is a cautionary tale, and an important one for anyone tempted to do half the job. Removing names but leaving pronouns in place does not merely work less well; it makes the male skew *worse*, climbing from 87.1% to 99.6% on stereotypically male roles and from 80.2% to 97.2% on neutral roles. With the name cue removed, the remaining pronoun cue carries even more weight and pushes the male document ahead almost everywhere. The lesson for deployers and regulators alike: a partial fix can be worse than no fix at all.

Why a careful-looking fix can still fail repays attention. A grammatically faithful neutralization—turning *he* into *they*, *him* into *them*, and *his* into *their*—cannot guarantee equal treatment, because English “*her*” does double duty: it is both the possessive (the counterpart of *his*) and the object (the counterpart of *him*). Mapping “*her*” to either one leaves the man’s and woman’s documents differing by a single word, and the re-ranker still picks up on it. A reliable transform therefore collapses *every* gendered pronoun to one and the same token, giving up grammatical polish in exchange for the guarantee that the two documents become exactly the same text.

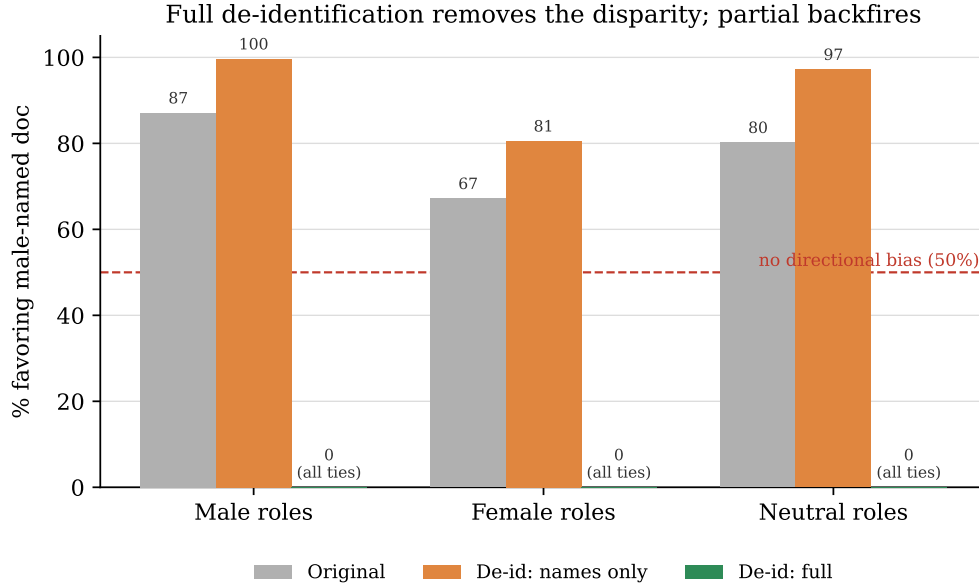


Figure 3: Share favoring the male-named document, by occupation category, under the original documents, a names-only transform, and the full transform. Removing names but leaving pronouns (*names only*) makes the male skew *worse*; the full transform makes each pair identical, so all pairs tie and the disparity is zero.

## 5 Legal Analysis

**Disparate impact is judged by the result, not the inner workings.** A Title VII disparate-impact claim does not turn on how a tool reaches its decision; it turns on whether a facially neutral practice produces a disproportionately adverse result for a protected class [2, 4]. The quantity that matters, then, is not the per-document score difference—which our own data show is small—but who actually gets selected. In our slate simulation the unit of selection is the shortlist, and its composition is lopsided. Under the full pipeline, women hold just 19% of top-3 spots overall, and 18% even for gender-neutral roles. Run through the Uniform Guidelines’ four-fifths rule, the female selection rate is roughly one-quarter of the male rate—far below the 0.80 floor that triggers an inference of adverse impact [11]. That the underlying score nudges are individually tiny is, as a matter of doctrine, beside the point. The Guidelines expressly allow that smaller disparities can be actionable when they are statistically and practically significant, and ranking does something the score gaps alone do not: it converts a string of small, consistent gaps into a hard line between who makes the cut and who does not.

**The “neutral occupation” finding is the sharpest.** For occupations that carry no stereotype, a non-discriminatory ranker should return a roughly balanced shortlist. Ours is 82% male. Crucially, there is no business-necessity story available here: one cannot argue that a gender skew is required by the job when the labor market itself does not gender the role. The skew is not a feature of the work; it is an artifact of the model. This is the cleanest disparate-impact posture in our data, because it strips away the employer’s usual justification before the argument even begins.

**De-identification is a textbook “less discriminatory alternative.”** Even where a practice might survive the business-necessity inquiry, liability attaches if a less discriminatory alternative exists and the employer declines to adopt it. De-identification is precisely such an alternative. It is cheap (a fixed set of substitution rules, with no model retraining); it is auditable (every substitution can be read and checked by hand); and, within the channels it covers, it is *provably* effective, driving the disparity to zero rather than merely reducing it. An employer who feeds gendered text into a ranker without de-identifying it—when de-identification is available at almost no cost—is in a weak position under this prong, and a plaintiff or regulator can point to a remedy that is concrete, tested, and already on the shelf.

**The EU AI Act points the same way.** The European Union reaches the same conclusion from the opposite direction. Under Regulation (EU) 2024/1689, AI systems used to filter job applications and evaluate candidates are classified as *high-risk*—the most stringent category short of an outright ban [10]. Strikingly, the Regulation’s own rationale names the precise harm we measure: it warns that such systems “may perpetuate historical patterns of discrimination, for example against women,” and may have “an appreciable impact on future career prospects, livelihoods [...] and workers’ rights” (Recital 57). Our experiment is, in effect, a direct measurement of the discrimination the Regulation singles out. High-risk status carries concrete duties—a continuous risk-management system for fundamental-rights risks (Art. 9), data governance that requires datasets to be examined for possible biases and equipped with measures to detect and mitigate them (Art. 10), and human oversight (Art. 14)—and de-identification is a cheap, documentable technical measure responsive to each. Because the harm compounds across the two stages, it must be applied at *both* the retrieval and re-ranking stages; covering only one leaves the duty unmet, because the bias simply re-enters at the other.

## 6 Policy Recommendation

We recommend that anyone deploying two-stage ranking over text that describes people de-identify gender markers in both the documents and the search queries *before* both the retrieval and the re-ranking stages. The specification below is deliberately minimal and fully auditable—a court, regulator, or in-house compliance team can read it, require it, and verify it:

1. **Normalize case.** Lowercase the text, so that capitalization cues (such as a sentence-initial “He” or “She”) cannot leak gender back in.
2. **Replace names.** Substitute personal names with a neutral placeholder, or delete them. The choice among placeholder, deletion, and blanking does not affect the fairness guarantee—both documents receive the same substitution either way—but it may affect how well the system still retrieves relevant candidates, and should be chosen with that trade-off in mind.
3. **Collapse gendered pronouns to a single token.** Replace every gendered pronoun with one uniform token. Do *not* settle for grammatical neutralization (*they/them/their*): as our names-only and grammatical-mapping results show, the “*her*” possessive/object collision leaves a residual gender signal the model can still detect.
4. **Apply at both stages, and audit the transform itself.** Because a biased re-ranker can erase a fair retriever’s output, covering only one stage is not enough. The list of substitutions



should be reviewed, and the transformed text spot-checked for any gendered terms that slipped through.

The guarantee rests on a simple symmetry. Once the only gender-differing words are removed, the man’s document and the woman’s document are identical, and *any* ranker—however biased its internals—must score identical documents equally. The fairness does not depend on trusting the model; it follows from the text.

## 7 Limitations

Several limits bound what we claim. Our remedy removes bias only on the channels it covers. Real résumés carry gender through signals that a name-and-pronoun rule does not touch—gendered organization and award names, single-sex school affiliations, hobbies, and even writing style. De-identifying names and pronouns is therefore *necessary but not sufficient* in the field; it is fully sufficient in our setting only because, by design, names and pronouns are the sole things that vary between the paired documents. Our compounding and de-identification experiments use a single representative pipeline (the `bge` stack), though the single-stage bias itself is documented across all fifteen re-rankers. Our documents are synthetic templates, simpler than real candidate text, and we do not measure how much retrieval accuracy is lost when names are replaced or blanked—a cost a deployer would need to weigh. Finally, we audit gender through binary male/female name and pronoun counterfactuals; we do not address non-binary identities or intersectional effects (such as the combined influence of gender and race), which prior work indicates can be larger still [13].

## 8 Conclusion

Re-rankers—the components that fix the final order a recruiter sees—carry a consistent gender preference that tracks occupational stereotypes. Run in sequence with a biased retriever, that disparity does not wash out. It concentrates in the shortlist, which is 81% male-named overall and 82% male-named even for gender-neutral roles, and a biased re-ranker can undo the work of a fairer retriever. The per-document magnitudes are small, but at the cutoff—the line between making the shortlist and not—they become legally consequential, and they map onto both U.S. disparate-impact doctrine and the EU AI Act’s high-risk regime. A deterministic de-identification built from a short set of substitution rules—lowercase the text, replace names, collapse gendered pronouns to one token—reduces the disparity to exactly zero, provided it is complete and applied at both stages. It is the rare fairness intervention that is at once cheap, auditable, and provable. We argue that those who deploy automated ranking over text describing people should be expected to adopt it.

**Data and code availability.** The counterfactual dataset generator, the fifteen-model scoring outputs, the pipeline simulation, and the de-identification transform are released with this paper.

Table A1: Share of counterfactual pairs in which the model favored the *male*-named document, by occupation category, for each of the fifteen re-rankers ( $n = 13,120$  pairs each); an even-handed model would sit at 50% in every column.

| Re-ranker                | Male jobs    | Female jobs  | Neutral      |
|--------------------------|--------------|--------------|--------------|
| rerank-2                 | 75.7%        | 8.9%         | 46.3%        |
| rerank-2.5               | 88.9%        | 15.9%        | 65.5%        |
| semantic-ranker-003      | 50.8%        | 10.1%        | 29.7%        |
| semantic-ranker-004      | 50.8%        | 10.1%        | 29.7%        |
| rerank-2-lite            | 39.0%        | 13.9%        | 29.0%        |
| semantic-ranker-fast-004 | 68.1%        | 21.4%        | 45.4%        |
| Cohere-rerank-v4.0-pro   | 65.2%        | 22.4%        | 49.9%        |
| rerank-2.5-lite          | 63.7%        | 27.5%        | 46.0%        |
| jina-reranker-v2         | 51.0%        | 27.6%        | 38.4%        |
| Cohere-rerank-v4.0-fast  | 63.2%        | 28.0%        | 57.1%        |
| rerank-1                 | 72.0%        | 29.0%        | 67.7%        |
| ms-marco-MiniLM-L-12     | 86.4%        | 47.4%        | 82.4%        |
| semantic-ranker-002      | 91.7%        | 50.7%        | 76.3%        |
| bge-reranker-v2-m3       | 87.1%        | 67.1%        | 80.2%        |
| ibm-granite-reranker-r2  | 90.6%        | 68.6%        | 86.2%        |
| <b>Mean</b>              | <b>69.6%</b> | <b>29.9%</b> | <b>55.3%</b> |

## A Per-model results

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